

Computer Vision Technologies in Movement Disorders: A Systematic Review

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Abstract: Background: Evaluation of movement disorders primarily relies on phenomenology. Despite refinements in diagnostic criteria, the accuracy remains suboptimal. Such a gap may be bridged by machine learning and video technology, which permit objective, quantitative, non-invasive motor analysis. Markerless automated video-analysis, namely Computer Vision, emerged as best suited for ecologically-valid assessment. Objectives: To systematically review the application of Computer Vision for assessment, diagnosis, and monitoring of movement disorders.

Methods: Following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines, we searched Cochrane, Embase, PubMed, and Scopus databases for articles published between 1984 and September 2024. We used the following search strategy: (“video analysis” OR “computer vision”) AND (“Parkinson’s disease” OR “PD” OR “tremor” OR “dystonia” OR “parkinsonism” OR “progressive supranuclear palsy” OR “PSP” OR “multiple system atrophy” OR “MSA” OR “corticobasal syndrome” OR “CBS” OR “chorea” OR “ballism” OR “myoclonus” OR “Tourette’s syndrome”).

Results: Out of 1099 identified studies, 61 met inclusion criteria, and 10 additional studies were included based on authors’ judgment. Parkinson’s disease was the most investigated movement disorder, with gait as the prevalent motor task. OpenPose was the most used pose estimation software. Automated video-analysis consistently achieved diagnostic accuracies exceeding 80% across most movement disorders. For tremor, dystonia severity and tic detection, Computer Vision strongly aligned with accelerometry and clinical assessments.

Conclusions: Computer Vision holds potential to provide non-invasive quantification of presence and severity of movement disorders. Heterogeneity in video settings, software usage, and definition of standardized guidelines for videorecording are challenges to be addressed for real-word applications.

To date, the diagnosis of a parkinsonian syndrome relies on cardinal motor symptoms, such as bradykinesia, associated with rest tremor and rigidity, variably combined.¹ Despite refinements, current diagnostic criteria for idiopathic Parkinson’s disease (PD) still result in a 20% error rate.^{2–5} Traditionally, the assessment of PD has relied heavily on clinical observation, using rating scales such as the Movement Disorders Society-Unified Parkinson’s Disease Rating Scale (MDS-UPDRS) and the

Hoehn & Yahr (H&Y) disability scale. These scales often fail to capture early and subtle motor symptoms, leading to potential diagnostic inaccuracies and delays. Subjectivity and inter-rater variability inherent in these scales present significant challenges, as even experienced clinicians may exhibit variability in scoring.⁶ Concerning hyperkinetic movement disorders, the diagnosis should rely on time features, space characteristics, and relationship with action/rest, suppressibility, wakefulness.^{7,8} In this

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regard, the revised Toronto Western Spasmodic Torticollis Rating Scale (TWSTRS-2) for cervical dystonia shows good overall inter- and intra-rater reliability in severity scoring, but certain items, such as sagittal shift, duration, and sensory trick, display poor inter-rater agreement.^{9–11}

The diagnostic uncertainty across movement disorders underscores the importance of reliable biomarkers, that represent a pressing, but still unmet need in clinical practice.

Traditional assessments typically occur in the short time of a clinical encounter that does not reflect patient's real-world conditions, and fluctuating symptoms, like freezing of gait (FoG) and levodopa-induced dyskinesias (LIDs), or nighttime axial rigidity, often go unnoticed.^{12,13} Clinical scales capture only episodic snapshots and fail to represent the dynamic nature of movement disorders, which can vary significantly even throughout the same day. These inconsistencies should pave the way for more ecologically-valid monitoring pipelines to cover the symptom spectrum in naturalistic contexts.^{13,14}

Videorecording is an essential component in the toolbox of movement disorders experts: it provides a unique record of physical examination, can capture nuances of movement over time and supplement the written medical record while being an indispensable resource for education.¹⁵ When the phenomenology is debated, videos offer the opportunity to re-review the movements of a patient who is no longer in the clinic.^{15,16} Advancements in artificial intelligence (AI) and machine learning (ML) models applied to video-captured frames have been fostering a more comprehensive characterization of movements, taking advantage of the commonly available user-friendly smartphone cameras.^{17,18}

Computer Vision (CV) hierarchically falls under both AI and ML,^{19,20} leveraging algorithms to interpret visual data, such as images and videos. CV uses the principles of ML to extract meaningful insights from visual information, serving as a bridge between raw visual data and actionable AI-driven outcomes. Video-based systems are categorized into marker-based and markerless technologies. Marker-based systems, while highly accurate, are labor-intensive and often require controlled environments, limiting their practical application to research settings.¹⁷ Markerless models, on the other hand, utilize CV to analyze movement without physical markers. The latter approach is more suited for continuous long-term monitoring, allowing comfortable evaluations while performing unconstrained activities at home.¹⁷ Evaluating limb bradykinesia using an ordinal severity scale for finger tapping is an oversimplification of complex movements. Identifying subtle variations in velocity, amplitude, and rhythm enriches the standard examination with a granular view of movements when compared to clinical scales. Figure 1 highlights the pipeline of CV for finger tapping, culminating in pattern classification (Panel A), and inertial and automated video-based models' workflow with a qualitative comparison²¹ (Panel B).

Pose estimation is the backbone of CV pipeline, as it allows identifying and tracking body coordinates, creating a skeletal model. For this purpose, landmark detection is fundamental, involving the identification of key points on the body to

facilitate movement analysis. These landmarks can be anatomical reference points, which are crucial for creating a skeletal model for movement recognition. Specifically, *OpenPose* and *MediaPipe* offer multi-person tracking and adaptability to various environments and are more suitable for gait, finger tapping and tremor analysis, while *Custom 3D* models focus on hand and finger tracking.^{17,22} *DeepLabCut* provides a frame-by-frame movement tracking, supporting both dystonia and bradykinesia evaluation.^{17,22,23} Table 1 lists applications of pose estimation software and applicable landmarks. Once skeletal data have been generated, the next step involves extracting quantitative kinematic features to capture joint relationships during motor tasks. Finally, extracted kinematic features feed ML models to classify movement (eg, abnormal or healthy) or predict clinical scores. For example, Support Vector Machine (SVM) is particularly useful to handle high-dimensional data during finger tapping, while Random Forest (RF) is more suitable when multiple motor symptoms are evaluated simultaneously, such as in home-based dyskinesia monitoring.

To date, no clinically oriented state-of-the-art overview about CV in movement disorders has been published, with most studies conducted from a bioengineering perspective.²² We here sought to redress this shortcoming.

Taking these premises into account, the present systematic review aims to unravel the current state of CV in movement disorders, disentangling the topic through video setups, current models, and key performance metrics in clinical tasks. Examining strengths and limitations of this emerging technology, we unveil its potential for augmenting disease diagnosis and monitoring, and provide practical recommendations for routine use of markerless video-analysis.

Methods

Search Strategy

This systematic review follows the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines.²⁴ We searched Cochrane, Embase, PubMed, and Scopus for articles published between January 1, 1984, and September 30, 2024, focusing on English language articles meeting inclusion criteria. The search string was:

- ("1984/01/01" [Date—Publication]: "2024/09/30" [Date—Publication]) AND (English [Language]) AND ("video analysis" OR "computer vision") AND ("Parkinson's disease" OR "PD" OR "tremor" OR "dystonia" OR "parkinsonism" OR "progressive supranuclear palsy" OR "PSP" OR "multiple system atrophy" OR "MSA" OR "corticobasal syndrome" OR "CBS" OR "chorea" OR "ballism" OR "myoclonus" OR "Tourette's syndrome")

The search strategies relative to each database, including technology and population terms, are reported in Table S1.

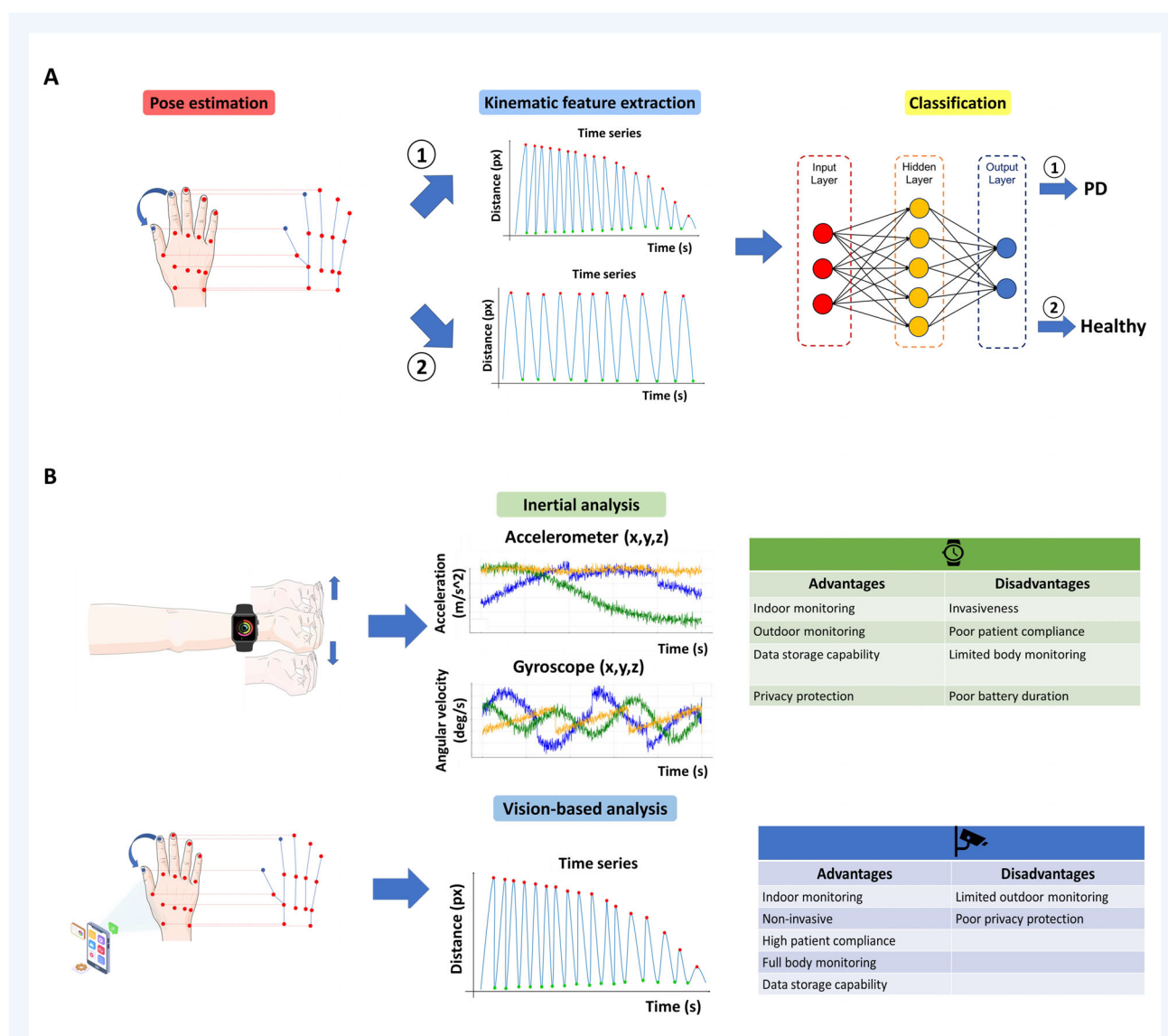


Figure 1. (A) A three-step process for finger tapping analysis using Computer Vision. First, key points on the hand and fingers are detected to create a skeletal representation, identifying the spatial coordinates of each joint. This phase captures the position and movement of individual fingers during tapping. The tip of the index and the thumb are chosen to analyze the finger tapping as the distance between these two points. In the second step, kinematic features (thumb-index distance) are extracted from the movement data, represented as waveforms over time. For 1, a progressive reduction in the thumb-index distance is observed, while in 2, the distance remains consistent over time, indicating different tapping patterns. Finally, these kinematic features are fed into a machine learning classifier, that categorizes the tapping pattern as either “pathological” or “healthy.” This classification step provides an objective definition based on the identified movement characteristics. (B) Workflow of both inertial and video-based models for quantitative movement analysis. Inertial Analysis (upper panel): A smartwatch, worn on the wrist of a patient with postural tremor when the arms are outstretched, collects acceleration and angular velocity data in the three axes (x, y, z) from accelerometer and gyroscope, respectively. Vision-based analysis (lower panel): A smartphone camera videotapes the finger tapping of a PD patient. The tip of the index and the thumb have been chosen to analyze the finger tapping as the distance between these two points and the model analyzes the variation in distance between the index and thumb tips over time. The authors suggest a qualitative comparison (right panels).

Eligibility Criteria

A title/abstract screening process was followed by a full-text screening. After screening of all studies, all raters merged their findings. No automation tool was used for the process. The eligibility criteria were designed according to the PICO (Population, Intervention, Comparison, Outcome) framework. The following PICO questions were investigated:

1. In studies on movement disorders, what is the performance of markerless CV-based technologies compared to traditional methods for disease diagnosis?
2. In studies on movement disorders, what is the performance of markerless CV-based technologies compared to traditional methods for quantitative phenotyping, severity assessment and monitoring?

To address these questions, eligibility criteria were defined based on the following parameters:

TABLE 1 Summary of main pose estimation models with the number of landmarks related to specific body district analysis and principal working features

Pose estimation	District (N landmarks)			Description
	Face	Hand	Full body	
Openpose	70	21/hand	25	Openpose accurately detects joint movements using skeletal landmarks across body regions, making it suitable for real-time gait analysis, postural assessment, and fine motor evaluations (eg, finger tapping and tremor). ^{109,110}
Mediapipe	468	21/hand	33	MediaPipe is a mobile-friendly tracking of human body landmarks, optimized for variable lighting and camera angles, ideal for unsupervised home-based assessments of upper limb function, facial expressivity, and gross motor tasks. ^{111,112}
Custom 3D hand pose estimation	NS	21/hand	NS	This specialized software provides precise 3D tracking of hand and finger movements, quantifying tapping velocity and amplitude, making it a valuable tool for assessing bradykinesia in PD. ¹¹³
DeepLabCut	C	C	C	DeepLabCut (DLC) offers a markerless approach to tracking specific movement trajectories, such as dystonic postures or compensatory motor strategies in gait. It also does not require specific positioning of the hand to provide accurate tracking of the digits. ²³

Abbreviations: C, customizable; N, number; NS, not supported.

- Population: Patients with movement disorders
- Intervention: Markerless CV-based technology for diagnosing, monitoring, or assessing the severity of movement disorders
- Comparison: Reference standards for validation, including clinical diagnosis, clinical rating scales, and sensor-based methods where applicable
- Outcome: Performance of CV-based assessments, measured by diagnostic performances (sensitivity, specificity, ROC AUC, accuracy), agreement with clinical assessments (intra-class correlation coefficient-ICC-with expert ratings), F1-score, precision, recall.

This PICO framework guided the selection of search terms, inclusion/exclusion criteria, data extraction, and result structuring in the review.

Two reviewers (P.M.P.; L.M.) independently screened literature, assessed each study and collected data. Reviewers considered only data that were reported in the published papers. Three senior authors (A.J.E.; M.B.; L.d.B.) were chosen to resolve conflicting results between reviewers.

A total of 1099 records were identified across the four databases (Cochrane, $n = 284$; Embase, $n = 312$; PubMed, $n = 164$; Scopus, $n = 339$). After removing duplicate records ($n = 283$), 816 unique records remained for screening. During the eligibility assessment, we excluded records that were retracted ($n = 2$), Letters to the Editor ($n = 3$), conference abstracts ($n = 23$), and non-English articles ($n = 1$), leaving 787 records for title and abstract screening. Of these, 653 were excluded based on relevance criteria, including articles that were preprints ($n = 1$) or in press ($n = 1$). A total of 134 full-text articles were assessed for eligibility, of which 73 were excluded due to the following reasons: studies not related to the selected movement disorders

($n = 27$), not focusing on video-based analysis ($n = 9$), not utilizing automated models ($n = 18$), not conducted on human subjects ($n = 2$), and review articles ($n = 17$). After completing the screening process, 61 studies met our inclusion criteria, and an additional 10 studies were included based on the authors' expertise and judgment, leading to a final selection of 71 studies. The selection process is visually represented in the PRISMA flowchart (Fig. 2).

Quality Assessment

The quality of the included studies was rigorously assessed using the Quality Assessment of Diagnostic Accuracy Studies (QUADAS-2) tool. This tool evaluated the risk of bias and applicability across four domains: patient selection, index test, reference standard, and the flow and timing of testing. All 71 included articles underwent a quality check to examine risk of bias and concerns regarding applicability to the review aim and PICO questions.²⁵

Results

Study Characteristics

Study Aims and Types of Movement Disorders

56.3% ($n = 40$) of studies focused on diagnostic purposes, while 39.4% ($n = 28$) and 4.2% ($n = 3$) of included works investigated disease severity and monitoring, respectively (Table S1). Most

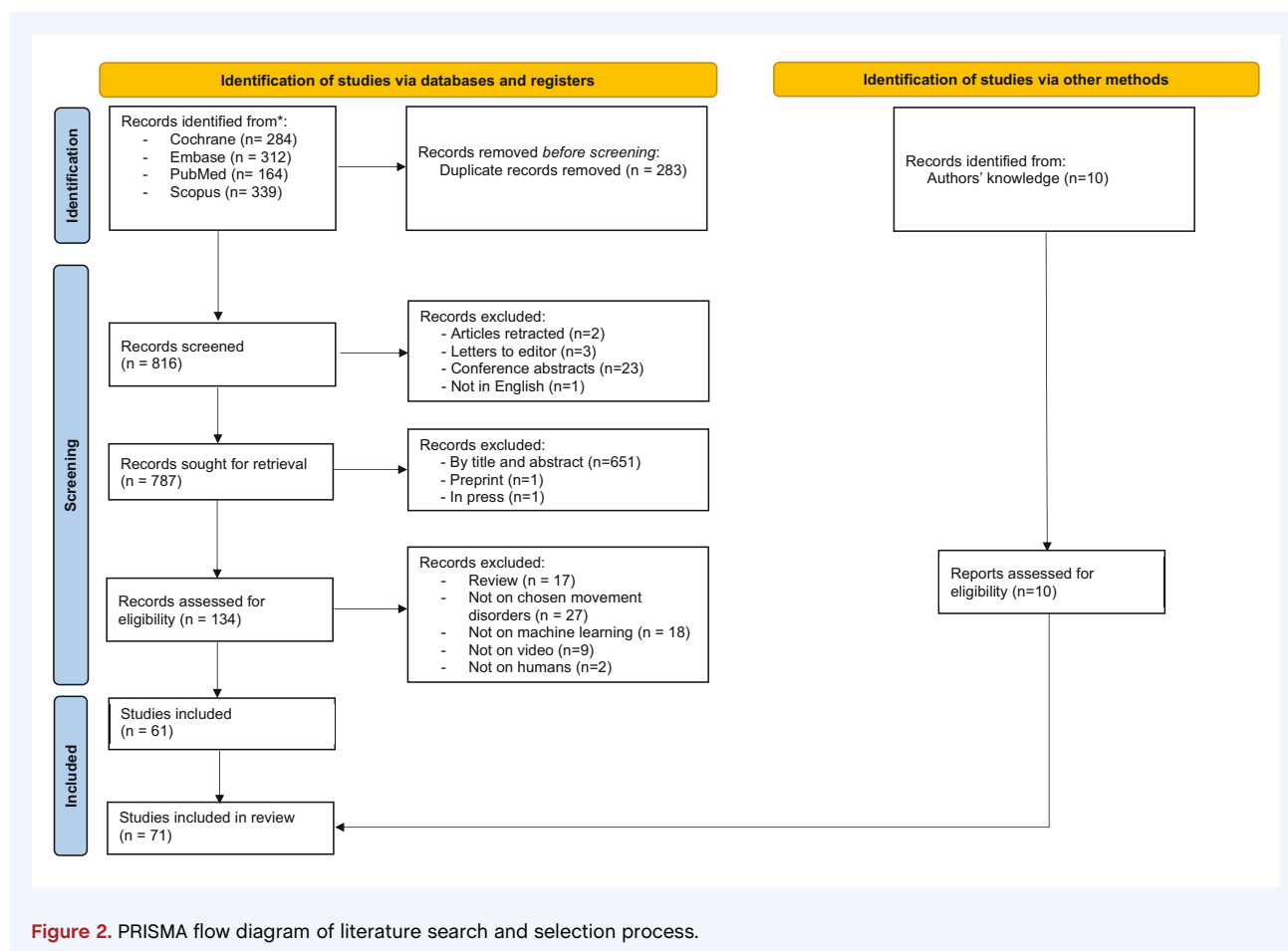


Figure 2. PRISMA flow diagram of literature search and selection process.

studies were PD-related ($n = 53$, 74.6%), followed by dystonia syndromes (cervical, generalized, blepharospasm-BSP, and adductor-type spasmodic dysphonia-ADSD) ($n = 9$, 12.7%), ET ($n = 4$, 5.6%) and hemifacial spasm (HFS) ($n = 2$, 3%). There was one study on Tourette's syndrome (TS) (1%). In two studies (3%), only healthy subjects (HS) mimicking movement disorders were enrolled (Table S2). No studies on atypical parkinsonism, chorea and ballism were found.

Clinical Features

Gait was targeted in most of the eligible studies ($n = 21$, 35%), of which 3% specifically on FoG. Others were limb or head tremor ($n = 10$, 14%) and finger tapping ($n = 10$, 14%), followed by facial expressions ($n = 5$, 7%), dystonia ($n = 9$, 13%), postural instability, abnormal axial postures, and LIDs ($n = 2$, or 3% each). In some studies, motor tasks were investigated in combination ($n = 4$, 6%), or through upper-limb kinematic analysis during deep brain stimulation (DBS) ($n = 3$, 4%) (Table S2).

Pose Estimation Models

OpenPose was the most frequently used software ($n = 16$, 23%), followed by MediaPipe ($n = 7$, 10%), DeepLabCut ($n = 5$, 7%),

and OpenCV ($n = 4$, 6%). Tools like AlphaPose, and Eulerian Magnification (EM), and other custom CV solutions were each used in 54% ($n = 38$) of studies (Table S2).

Figure 3 summarizes explored movement disorders (top panel, A), symptoms (middle panel, B), and relative representation of CV software employed (lower panel, C).

Clinical Investigations

Based on tasks investigated, enrolled subjects, recording devices, CV software, and performance metrics (Table S2), we first categorized the findings according to PICO questions: diagnosis, severity assessment, and monitoring. Within each category, studies were further classified based on the investigated movement disorder, and subsequently on the tasks assessed. Concerning PD, we presented results for clinical tasks^{26–64} (finger tapping, combined motor tasks, and gait) and non-tasks^{40,65–84} (face expressions, tremor, posture and balance, LIDs, and dystonia). Additionally, some studies showcased the usefulness of CV in dystonia syndromes,^{80,84–92} HFS,^{93,94} ET^{72,75} and TS.⁹⁵ To ensure consistency, we focused exclusively on performance metrics reported in each study.

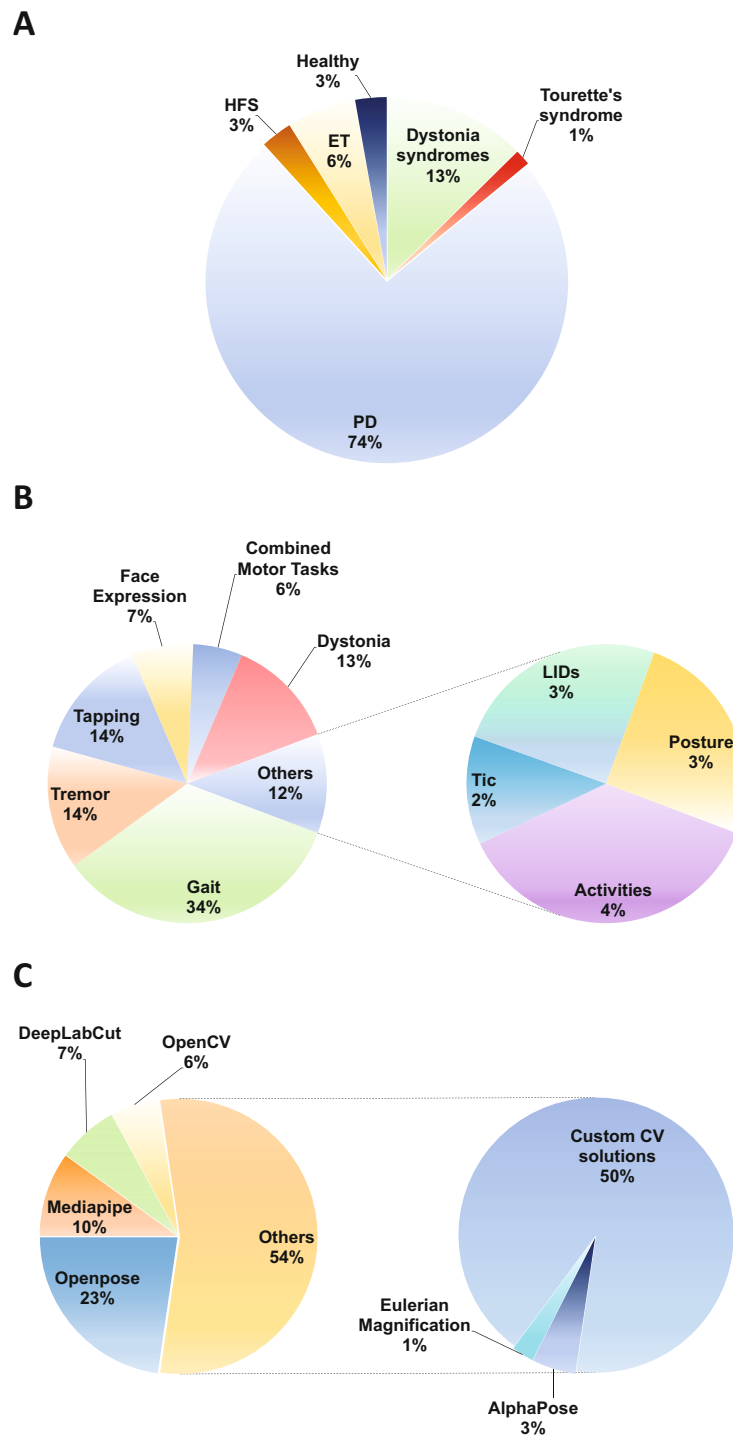


Figure 3. Findings from application of the search strategy to the review question. (A) Percentage of movement disorders investigated in analyzed studies. (B) Percentage of motor tasks investigated in search studies. (C) Relative representation of Computer Vision software used in literature studies. ET, Essential Tremor; HFS, Hemifacial spasm; LIDs, Levodopa-induced dyskinesias; PD, Parkinson's disease.

Diagnosis

PD-Clinical Tasks

Finger Tapping. A 3D convolutional neural network was applied directly to video data without manual feature extraction, showing an AUC of 0.76 in differentiating PD from controls.²⁸ In another framework, spatial-temporal 3D hand pose estimation analyzed videos captured with a depth camera, attaining a classification accuracy of 81.2% for finger tapping.³²

Combined Motor Tasks. HandSegNet and PoseNet were deployed to localize hand key points in video frames, achieving an accuracy rate of 87% in detecting global bradykinesia during hand movements.³⁴ 42 PD patients, 24 HS, and 13 with ET performed 15 motor tasks while data were collected using inertial sensors and a Logitech BRIO 4 K camera. The extracted features were analyzed using RF and XGBoost classifiers to differentiate between PD vs HS and PD vs ET. Results demonstrated that combining sensor and video data improved classification performance, with an F1-score increase of up to 18% for PD detection and 21% for distinguishing PD from ET.⁷⁰

Gait. Deep learning features were combined with kinematic parameters obtained using optical flow, resulting in an overall accuracy of 92.3% for detecting PD-specific gait impairments.⁴³ Notably, this approach achieved 100% accuracy when integrating gait and eye fixation modalities to discriminate between patients and HS. Another study proposed smartphone videos and human pose estimation to extract three-dimensional joint coordinates, subsequently converted into color-mapped images for classification using RF models, achieving an accuracy of 91.67%.⁴⁴

PD-Non Tasks

Facial Expressions. A total of 106 facial landmarks were extracted from short videos of facial expressions to quantify subtle movements of 33 PD patients and 31 HS.⁶⁶ Authors employed a Face++ pose estimation model, achieving an impressive F1 score of 99% in distinguishing PD patients from HS.

Tremor. EM was employed to enhance the visibility of subclinical tremors in smartphone videos, significantly improving clinician detection rates with an odds ratio of 2.67,⁷³ while an OpenPose-based model differentiated PD and ET, achieving an F1 score of 0.84.³⁶

Posture and Balance. A CV-based model assessed the ability of PD patients to rise from a chair (MDS-UPDRS-III item 3.9) and demonstrated a sensitivity of 62.8% for detecting motor impairment and a specificity of 90.3% for identifying HS.⁷⁹

LIDs. Convolutional Pose Machines extracted movement trajectories from videos of 9 PD patients on levodopa-carbidopa intestinal gel infusion.⁸³ With RF models, authors predicted the severity of LIDs with high accuracy during communication tasks (AUC 0.93), as describing an image and engaging in discussion

with the examiner, while toe tapping was better for LIDs detection (AUC 0.773).

Dystonia. Head pose motion abnormalities were analyzed in 199 PD patients and 130 HS using an intelligent web-based assessment tool and OpenFace software. PD patients exhibited significantly higher random side-to-side head motion ($p = 0.001$, $d = 0.13$). Among 151 PD patients, motion randomness decreased exponentially with time since the last dose ($r = -0.404$, $p < 0.001$), suggesting a relationship with LIDs/dystonia.⁸⁴

Essential Tremor

The TremAn software has been validated as a reliable tool for the automatic assessment of tremor frequency from videos of ET patients. This software demonstrated a high concordance with accelerometry, showing a frequency difference of less than 0.1 Hz in 80% of the analyzed samples and less than 0.5 Hz in 94% of the samples.⁷²

Tourette's Syndrome

A single work validated two different automated models to detect and classify motor tics in 35 patients with TS from videos. The authors employed manually extracted features from MediaPipe Face Mesh landmarks and a deep neural network for facial landmark extraction. Both models achieved high accuracy (88.4% and 88.5%, respectively).⁹⁵

Severity Assessment

PD-Clinical Tasks

Finger Tapping. DeepLabCut processed smartphone videos of finger tapping, achieving high correlations with bradykinesia clinical ratings (Spearman coefficients up to -0.74 for speed and -0.66 for amplitude).²⁷ DeepLabCut and HandGraphCNN were integrated to create 2D and 3D skeletal hand models from smartphone videos, finding significant correlations with clinician-assessed bradykinesia scores (R^2 values of 0.609 for finger tapping).²⁹

Combined Motor Tasks. A novel CV activity recognition framework using OpenPose for pose estimation and spatial-temporal graphs processed by deep-learning models demonstrated performance comparable to clinical ratings in classifying 14 motor activities from the MDS-UPDRS-III.³³ Unstructured smartphone videos were pose-estimated by PIXIE, achieving an F1 score of 0.93 in classifying symptoms during finger-to-nose tasks.³⁸ Finally, DeepLabCut was proven to extract kinematic data during DBS implantation, achieving an overall classification accuracy of 85.7% for different upper-limb movements.⁹⁶

Gait. A markerless 2D video system employing a mixture of Gaussians estimated temporal PD gait features, achieving an ICC of approximately 0.85 compared to an optoelectronic system.⁴¹

A graph convolutional neural network represented body joints as directed graphs, improving the detection of FoG episodes with an AUC of 0.887.⁶⁴

PD-Non Tasks

Face Expressions. A pilot study used a deep convolutional neural network to detect hypomimia cues from facial expressions in PD patients.⁶⁷ Analyzing videos from 107 PD patients in both ON and OFF medication states and 1595 controls, their model achieved an AUC of 0.71 for recognizing hypomimia. Additionally, the model demonstrated a 63% accuracy in classifying medication states, compared to a clinical rater accuracy of 46%.

Tremor. OpenPose integrated to a Global Temporal-difference Shift Network evaluated tremor severity in PD, achieving accuracies up to 90.6% for hand resting tremor.⁷¹ Finally, smartphone-based methods also demonstrated excellent agreement with accelerometry for tremor frequency, with 97% of video measurements within 0.5 Hz of the clinical gold standard.⁷⁶

Posture and Balance. A camera- and viewpoint-agnostic approach evaluated camptocormia and Pisa syndrome in 55 PD patients using augmented human pose estimation (AutoPosturePD) to extrapolate specific axial landmarks.⁸⁰ OpenPose proved robust performances in extracting canonical and additional body keypoints for posture assessment across different imaging settings, making it feasible for clinical use without specialized equipment or trained personnel. Another CV-based model assessed patients' ability to rise from a chair and achieved an exact match with clinician ratings in 79.2% of cases and was within one point in 100% of cases.⁷⁹ For postural stability, remote scoring models were developed for rigidity and postural instability in 104 PD patients using low-cost RGB cameras.⁸¹ Features were extracted using OpenPose and combined with LightGBM algorithm, achieving moderate-to-substantial agreement with clinical scores (weighted kappa 0.58–0.66) and accuracy 70–73%.

Essential Tremor

In a cohort of ET patients, Mediapipe demonstrated convergent clinical validity with scores (Spearman's $\rho = 0.55$ – 0.86 , $p \leq 0.01$) and an accuracy for tremor amplitude and frequency comparable to optoelectronic marker-based systems, with a mean absolute error of up to 2.60 mm for amplitude and ≤ 0.21 Hz for frequency.⁷⁵ In another study, the accuracy of CV-based for resting and postural tremor frequency analysis from smartphone video recordings was compared to accelerometry in nine patients with PD, five with ET, and one with functional tremor. The methodology used optical flow-based movement tracking to extract pixel motion, with frequency analysis performed using Fast Fourier Transform (FFT). Bland–Altman analysis showed excellent agreement between video-derived and accelerometer-based tremor frequency, with 95% limits of agreement ranging from

–0.38 Hz to +0.35 Hz. In 97% of cases, the difference between methods was <0.5 Hz.⁷⁶

Dystonia Syndromes

The Computational Motor Objective Rater quantified head posture in 185 cervical dystonia patients, showing strong correlations with TWSTRS scores ($\rho = 0.59$ – 0.68).⁸⁵ It also assessed tremor component in other 93 patients, correlating with clinical ratings ($\rho = 0.54$, $p < 0.001$) and identifying primary tremor axes (50% pitch, 44% yaw, 6% roll).⁸⁹ DeepLabCut enabled markerless video analysis of dystonia in cerebral palsy, predicting arm and leg dystonia scores with errors comparable to inter-rater accuracy.⁸⁶ A MediaPipe-based model analyzed head movement in cervical dystonia, identifying features correlating with severity and subtypes.⁸⁸ For BSP, neural network-based software classified spasms in nine patients with high sensitivity for brief and prolonged spasms but lower sensitivity for blinking. CERT software showed moderate correlations with clinical scales ($\rho = 0.52$ – 0.56 , $p < 0.0001$), supporting its use for automated severity assessment.⁹² In ADSD, a CV model analyzed glottal geometry in nasolaryngoscopic videos, explaining 30–34% of clinical rating variance ($\rho = 0.55$ – 0.61 , $p < 0.001$).⁹¹

Hemifacial Spasm

A total of 39 patients (12 BSP, and 27 HFS) were recorded by Osaki and colleagues⁹³ using a high-speed camera, with eyelid movement energy quantified through FFT analysis before and after botulinum toxin injection. Significant post-treatment reductions in the total power spectrum of eyelid movements were observed for both BSP (from 109,213.6 to 55,661.1 mm²/s, $p = 0.0018$) and HFS (from 43,551.7 to 20,878.4 mm²/s, $p = 0.0058$).⁹³ In another work,⁹⁴ 22 HFS patients were similarly assessed, showing a significant decrease in energy on the affected side (from 46,999.5 to 21,463.8 mm²/s, $p = 0.0022$), with strong correlations between energy values and clinical severity as measured by Jankovic Rating Scale and Hemifacial Spasm Grading System.⁹⁴

Monitoring

PD

Combined Activities. Automated video-based analysis of sit-to-stand (STS) transitions to monitor 24 PD patients in a real-world setting over 85 h using wall-mounted cameras and OpenPose software to extract STS parameters.³⁷ STS duration and speed correlated significantly with MDS-UPDRS-III and postural instability and gait scores. PD patients had slower STS transitions than controls ($p < 0.001$) and further slowing OFF medication ($p = 0.016$). MCPD-Net, a multimodal deep learning model, leveraged automated silhouette analysis of 10 PD patients while performing unscripted cooking activities at home. MCPD-Net achieved an F1-score improvement of 0.25 (unimodal approach) and 0.09 (multimodal approach).³⁹ An

FPGA-based tracking system for scalable real-time monitoring of PD patients was tested using Gaussian Mixture Model (GMM) and Kalman filtering on a ZedBoard to enhance efficiency and privacy.⁴⁰ Results showed a precision ranging from 39.91% and 79.83% with recall from 6.04% and 18.13%.

Quality Appraisal

Authors substantially agreed that included studies had unclear risk of bias for patient selection (55% unclear, 45% low), reference standard (55% unclear, 45% low), and patient flow and intervention timing (63.3% unclear, 36.6% low), while the risk of bias was low only for index test (94% low, 6% unclear). Nonetheless, the authors identified for the majority of works low concerns regarding applicability to the research question for patient selection (21.2% unclear, 78.8% low) and reference standard (22.6% unclear and 77.4% low) domains and concordantly low concerns regarding applicability for index test (100% low). A checklist of quality metrics is summarized in Table S3, while Figure S1 depicts a graphical representation of mean values of all Authors' scores, from low to high, of QUADAS-2 items for risk of bias and concerns regarding applicability to the review question according to Authors' judgment.

Main Study Findings

Regarding the key findings of our PICO questions, diagnosis emerged as the primary aim (58%), whereas disease monitoring was the least investigated (4.3%). Gait was deeply analyzed, showing consistently high-performance standards across different setups. Applications of CV for finger tapping, tremor, and gait in PD delivered robust accuracies often exceeding 80%, prompting the strong capability to correctly recognize parkinsonian motor symptoms and extract key features from videos. Introducing CV also in ET and dystonia showcased comparable advancements, with pose estimation software strongly aligning with accelerometry for tremor and clinical ratings for dystonia. This technology shows strong potential for disease tracking and symptom monitoring in naturalistic, home-based settings, highlighting the underrated versatility of CV in capturing nuanced movement features across diverse environments (Table S2).^{37,39,40}

Noteworthy, studies focusing on PD accounted for 74.6% of the included works, with no one on atypical parkinsonism, chorea, ballism, and myoclonus other than HFS. This "PD-centric" view reflects both a limited application of CV in other conditions within the realm of movement disorders and atypical parkinsonism, as well as a limitation in our search string, which purposely excluded conditions like ataxias, neuropathies, and dementias. We prioritized specificity and homogeneity of the search to ensure methodological consistency. Including a broader range of disorders, such as functional movement disorders, would have introduced significant heterogeneity in movement phenomenology, and assessment methods, complicating the interpretation and synthesis of results. Moreover, CV applications in some of these conditions remain in early exploratory stages, which could have affected the robustness of our systematic

review. By narrowing the scope, we aimed to provide a more clinically relevant evaluation of CV technologies in movement disorders.

Clinical Relevance: The Need for Digital Biomarkers

From a clinical standpoint, scales like MDS-UPDRS-III, despite ease of accessibility to training are prone to high inter-rater variability.^{97,98} Indeed, caution should be warranted when interpreting high ICC values: to quantify a scale's ability to assess changes over time, a measure of reliability that incorporates the variance of the within-subject changes is mandatory.⁹⁸ The within-subject reliability of MDS-UPDRS-III has been proven to range between 0.13 and 0.62, confirming a significant amount of error variance.⁹⁸ In addition, in clinical practice, the imprecise use of *bradykinesia*, with reduction in amplitude and pauses in movements (hesitations/halts) as interchangeable, introduces ambiguity into ratings: patients can have *bradykinesia* without evidence of sequence effect.^{99–101}

We recognize three clear-cut steps forward in leveraging CV in the movement disorders clinic of the near future.^{102,103} A first takeaway is the exclusion of wearables, which may be uncomfortable for prolonged monitoring in real-life conditions. Limited data resolution, inconsistent interoperability, data quality and noise, periodic battery charging, and calibration issues, as sensor displacement or signal loss, are challenges to be addressed for a real-world usage.^{13,18,20,104–106} Nonetheless, seemingly objective home-collected data may lack reliability for motor characterization without a robust statistical analysis,¹⁰⁵ and differentiating bradykinesia from fatigue or fear of falling only with sensors can be misleading. The user-friendly nature of video-suited technology enhances patient compliance and allows for frequent assessments simply using a common and widely commercialized smartphone.^{15,17,18} On the counterpoint, wearable-developing companies must navigate regulatory approvals, manage intellectual property, and ensure successful commercialization. Indeed, only 6% of wearable technologies for PD have reached Technology Readiness Level (TRL) 9, indicating proof of functionality in operational environments¹⁰⁷ (Fig. 1). However, video-analysis heavily relies on stable camera positioning, consistent lighting conditions, and an unobstructed field of view—factors that are difficult to control in outdoor settings. Environmental variability, such as changes in lighting, weather conditions, or background movements, can introduce noise and reduce the accuracy of pose estimation algorithms (Fig. 1). On this topic, a video-based guide for remote (at-home) MDS-UPDRS-III administration has been recently validated, proving usability scores above the benchmark for effectiveness.¹⁰⁸ Second, dissecting movements quantitatively identifies subtle abnormalities invisible to physician's eyes. A recent non-automated blind-rated evaluation of speed, amplitude, rhythm, and sequence effect in finger-tapping videos of pathology-proven parkinsonian (PD and non-PD) patients revealed no significant differences between synucleinopathies and tauopathies.¹⁰⁹ Finally, a CV-assisted evaluation, as a supporting

tool for clinicians, could be crucial in clinical trials. Augmenting the level of sensitivity and statistical power, with reliable outcome measures is critical in patient-centered research, where digital biomarkers for disease progression tracking have become a pressing need to detect even small therapeutic benefits.

Notably, a search on ClinicalTrials.gov using the same search strategy revealed no ongoing clinical trials involving CV as a tool for digital biomarkers of disease progression or treatment response; conversely, most ongoing trials still rely on MDS-UPDRS subitems or, in some cases, on inertial data from wearable sensors.¹¹⁰

Unlike a previous systematic review on the topic,²² which primarily focused on deep learning for 2D/3D automated video-analysis, we adopted a clinically-oriented approach, emphasizing practical applications of CV models in real-world clinical settings. While Tang and colleagues²² extensively discussed cornerstones as algorithmic architectures and data modalities, our review provides insights into the external validity, feasibility, generalizability, and potential barriers to the integration of CV technologies in healthcare practice. This perspective is intended to guide clinicians and researchers in selecting tailored CV approaches for specific movement disorders and clinical objectives.

Based on the current evidence, we recommend prioritizing pose estimation models like OpenPose and MediaPipe, due to their robust accuracy, ease of integration, and adaptability across diverse clinical environments. OpenPose demonstrates superior performance in tasks requiring detailed multi-joint analysis, such as gait assessment and finger tapping, while MediaPipe offers high efficiency for real-time monitoring applications, including tremor detection and postural stability. For dystonia and tremor, DeepLabCut provides advanced tracking capabilities with frame-by-frame movement assessment, making it ideal for research-focused applications and specialized clinical settings. Conversely, for routine clinical use, frameworks with simpler implementation and lower computational demands, like MediaPipe, seem preferable. When choosing CV, clinicians should consider the clinical objective: it is essential to adopt a strategy tailored on the specific movement disorder and intended analysis. For example, the same movement disorder (eg, dystonia) may require distinct methodologies if the clinical aim changes (diagnosis vs severity assessment), emphasizing the need to align the choice of CV tools with the motor features under investigation and the desired outcomes.

Limitations

Considerable heterogeneity in study designs remains a major challenge in the current literature on CV in movement disorders. Methodological inconsistencies introduce assessment variability and complicate direct comparisons across studies. At present, there is no universally accepted reference protocol for video acquisition, leading to significant variability in camera placement, resolution, frame rate, lighting, and other environmental conditions. This lack of standardization limits the comparability and reproducibility of findings, and underscores the urgent need for shared guidelines in the field. Our quality assessment highlighted

notable risk of bias, mainly due to incomplete reporting of key methodological details. Diagnostic criteria were often unspecified, and information on disease duration and severity was frequently missing, limiting the interpretability of results across disease stages. In contrast, concerns regarding applicability were limited, with only ~20% of studies rated as “unclear” for patient selection and reference standard, reflecting a good methodological alignment with our PICO framework. However, QUADAS-2 does not formally assess the real-world applicability, which remains a relevant issue. Practical barriers such as equipment availability, controlled acquisition requirements, and the lack of standardized protocols may limit implementation of CV approaches in the clinic, and should be considered when interpreting their generalizability and utility.

Future Directions

The proliferation of technologies applied to movement disorders is driving a transformative process of *medical democratization*.¹³ This evolution allows healthcare providers to deliver equitable services, empowering patients to actively manage their health, while potentially reducing overall healthcare expenses. Integrating smart technologies into clinical practice improves decision-making, easing financial strain and addressing resource shrinking.¹³ To foster the practical utility of markerless video-based technology, clearly defined use-cases are mandatory: proliferation of models leads to a cluttered landscape of advanced technologies with unclear ecological/clinical relevance, potentially failing to align with patients' real needs (*ecological validity* pitfall). While valuable, large volumes of video-derived data should complement, not replace, deep phenotyping to avoid the pitfalls of overreliance on big data (*big data* pitfall).

To fully leverage the potential of CV in movement disorders and enhance reproducibility, comparability and generalizability across studies, collection and processing steps must be standardized for each use-case.¹⁷ For instance, if the hand is obscured during finger tapping, the pose estimation algorithm may fail to track landmarks, leading to errors in the final classification. Table S4 lists practical recommendations to improve data quality, ensuring consistency, and enhance model performance. Additionally, CV models are trained on relatively homogeneous datasets, which can limit generalizability to diverse patient populations. For example, models trained on younger, healthier individuals may not perform as well when applied to elderly patients with more severe impairments. External validation is paramount to ensure models perform effectively not only on training data but also on new datasets with different underlying distributions. Addressing this issue requires more diverse datasets and the use of transfer learning techniques to adapt models to populations with different phenotypes and disease severities.^{105,106} Regular retraining with updated labeled data helps preventing non-accurate predictions and underfitting due to data patterns shifts:¹⁰⁵ promoting open-source initiatives that engage both healthy individuals and patients through user-friendly mobile apps can facilitate the collection of video data during daily activities, enhancing model robustness. Personalizing informed consent, ensuring data anonymization, and implementing data

TABLE 2 Key issues in Computer Vision technologies for movement disorders: current state and proposed solutions

Problem	Current state	Possible solution
Inconsistent video data collection and setup	Variations in videorecording protocols, lighting and setup reduce accuracy	Standardize video capture and processing protocols (Table S3)
Low model generalizability	Models perform poorly on diverse populations	Use more comprehensive datasets and apply transfer learning ^a to adapt to diverse demographics and impairments
Lack of model transparency	Black-box models are hard to interpret	Develop interpretable models aligned with clinical intuition, integrating XAI approaches
Privacy and ethical issues in video data use	Weak standard consent, anonymization, and security measures	Strengthen consent, anonymization, and security through personalized consent forms and robust encryption
Non-motor symptoms overlooked	Autonomic, sleep and cognitive symptoms assessed via recall	Integrate non-motor symptoms' monitoring in naturalistic conditions
Limited assessment of fluctuating movement disorders	Evaluation limited by short encounters	Remote monitoring in ecologically-valid environments
Poor device interoperability	Fragmented device ecosystems	Create interoperable, multimodal platforms
Absence of normative cut-offs	Large amount of collected data lack direct clinical applicability	Promote open-source mobile apps for video data collection; establish benchmark datasets with normative ranges
Difficulty in addressing non-linear disease progression with poor real-world applicability	Current data lacks temporal and longitudinal context	Conduct large-scale, studies with tailored endpoints for real-world clinical and ecological application

^aA machine learning technique where a model developed for a specific task is repurposed or fine-tuned for a different but related task, leveraging knowledge gained from the initial task to improve performance and reduce the need for extensive data in the target domain.

Abbreviation: XAI, explainable Artificial Intelligence.

security measures, as encryption, are crucial ethical considerations for fostering patient trust and guaranteeing equitable access to these emerging technologies. While striking, deep learning models are often described as “black box” because it is difficult to interpret how they arrive at their conclusions, posing a trade-off between interpretability and complexity. In clinical settings, this lack of transparency can hinder trust in the decision-making process. There is a growing interest in developing more interpretable models, where the output is more transparent and aligns with clinical intuition^{13,17,105} and ensures fairness of data-driven prediction tools.

Large-scale studies are necessary to validate the long-term reliability of this technology, but endpoints need to be tailored for real-world applications. This may necessitate more nuanced outcomes, with longer data collection periods to develop a benchmark video-dataset, encompassing both normative and pathological data (Table 2).

Nonetheless, while motor symptoms remain the primary focus of periodic assessment, core non-motor features significantly impact patients' quality of life, even more than their motor counterpart, but are often overlooked in traditional encounters, delaying adjustments to treatment.^{13,14} Long-term monitoring of non-motor symptoms, as opposed to relying solely on patient

recall during visits, offers significant improvements in therapeutic outcomes.^{13,14} Interoperable platforms allowing for seamless data integration across different devices and healthcare systems will be crucial in the future: incorporating multimodal platforms, that combine data from various sources, into routine care facilitates ongoing Telemedicine trends and improves patient-reported outcomes. In the future, enabling interoperability across devices from different manufacturers (eg, combining data from wearable, biochemical, camera, and radar devices) will introduce more comprehensive datasets, improving decision-making processes and facilitating holistic, patient-centered care.^{13,20,111} The goal is a discrete unobtrusive monitoring fostering a personalized approach in movement disorder care.

Conclusion

This systematic review overviews the state of the art on markerless video-based technology as effective non-invasive tool for a quantitative assessment of movement disorders. While promising, challenges must still be faced to bridge the translational gap between research contexts and routine application of CV in the

clinic, including heterogeneity in video settings, variability in software, and the need for standardized data collection guidelines. Addressing these challenges promotes generalizability and real-world application, offering more accurate and accessible options for routine practice.

Author Roles

(1) Research project: A. Conception, B. Organization, C. Execution; (2) Statistical Analysis: A. Design, B. Execution, C. Review and Critique; (3) Manuscript Preparation: A. Writing of the first draft, B. Review and Critique.

P.M.P.: 1B, 1C, 2A,2B, 3A.

L.M.: 1B, 2C, 3A.

L.d.B.: 1A, 1B, 2C, 3B.

A.J.E.: 3B.

M.B.: 1A, 1B, 3B.

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Data Availability Statement

The data used to support this study's findings are available from the corresponding author upon reasonable request. ■

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Supporting Information

Supporting information may be found in the online version of this article.

Figure S1. Graphical representation of mean values of all five authors' scores of QUADAS-2 tool items for risk of bias and concerns regarding applicability to the review question.

Table S1. Search strings used for each database, including technology and population terms, with language and publication year filters.

Table S2. Summary of literature of studies focusing on Computer Vision classified per patients and controls enrolled, pose estimation software, recording device and performance.

Table S3. Quality metrics of included studies according to QUADAS-2 score with each cell containing assessments from each Author.

Table S4. Practical recommendation for videotaping movement disorders in the clinic.